# Passive buzzer

DO PICO SOUND PWM

## What does it do?

A passive buzzer needs fast pulses (PWM) to make sound. Changing their frequency changes its pitch: low or high notes.

**Recognise it:** check for **passive**, **external drive** or **no oscillator** on the packet. Both types can look identical; + only marks polarity.

Use a small **3.3 V GPIO-compatible piezo** sounder.

## Uses



### Washing-machine chime

PWM plays tunes; active buzzers have only one pitch.

**DETECT** Read cycle status → **DECIDE** Has the wash finished? → **DO** Play a short tune



### Robot voice box

Change pitch for happy chirps or grumpy growls.

**DETECT** Read mood button → **DECIDE** Is happy pressed? → **DO** Chirp high or low



**+ leg**
Marked +, to GPIO

**− leg**
Other leg, to GND

**Vent**
Sound comes out



Raspberry Pi Foundation

# Code

rpf.io/soundfx

## Raspberry Pi Pico

```
from picozero import PWMBuzzer

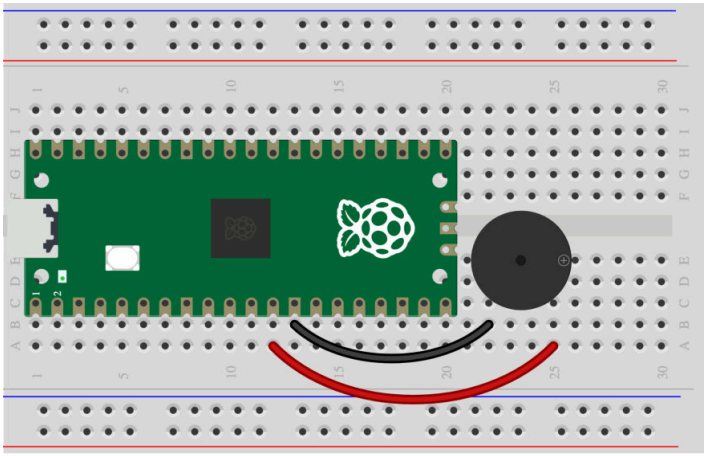
# GP9, 2000 Hz, 50% duty
buzzer = PWMBuzzer(9, 2000, 32768)

buzzer.on()
```

Try `buzzer.freq = 1000`; silence with `buzzer.off()`. Higher frequency means higher pitch. Beep speed is separate; both types can make rhythms.

# Wiring

**PICO**



**GP9** (pin 12) → **+** buzzer **–** → **GND** (pin 13)